

Project 5: Customer Support RAG-Powered Intelligent Chatbot

Project Overview

Build a production-ready support automation system using **Retrieval-Augmented Generation (RAG)** and vector search. It should serve real queries and integrate with business backend systems.

Milestone 1: Data Collection & Preprocessing

Objectives

- Aggregate support ticket logs, FAQs, documentation, and knowledge base

Tasks

1. Data Ingestion:

- a. Collect historical support tickets, product manuals, and FAQ entries.

2. Preprocessing:

- a. Clean and unify text data, remove noise, and tokenize.

3. EDA:

- a. Analyze common queries, topic distributions, and response effectiveness.

Deliverables

- Processed Text Corpus
- Preprocessing Pipeline Doc
- Support Data EDA Report

Milestone 2: Model Development & Evaluation

Objectives

- Train and evaluate RAG retrieval + generation pipeline

Tasks

1. Model Building:

- a. Create vector store (e.g., Azure Cognitive Search or embeddings + vector DB).
- b. Configure RAG model using Hugging Face or similar.

2. Evaluation:

- a. Evaluate with metrics like BLEU, ROUGE, and relevance scoring.

3. Optimization:

- a. Fine-tune RAG components based on relevance and response quality.

Deliverables

- Trained RAG Pipeline
- Model Evaluation Report

Milestone 3: Advanced Techniques & Azure Deployment

Objectives

- Deploy intelligent chatbot as part of production support

Tasks

1. Azure Integration:

- a. Deploy the RAG service via Azure Machine Learning or Azure App Service.

2. API & Workflow Integration:

- a. Create REST API and integrate with support portal.

3. Security & Access:

- a. Secure endpoints with Azure AD or API keys.

Deliverables

- Deployed Chatbot Service
- Integration & Security Doc

Milestone 4: MLOps & Monitoring

Objectives

- Track performance and automate retraining

Tasks

1. Experiment Tracking:

- a. Use MLflow tracking for RAG variations.

2. Monitoring:

- a. Monitor accuracy, latency, user satisfaction scores.

3. Retraining Mechanism:

- a. Schedule refresh of embeddings and model weights.

Deliverables

- Monitoring Dashboard
- Retraining Pipeline

Milestone 5: Final Documentation & Presentation

Deliverables

- Final Report
- Demo Presentation
- Business KPI Impact Analysis